

GBFsen: The Speed of Urgency

Giovanni was thirty years old, a lawyer in southern Italy, when the diagnosis finally gave a name to what his family had feared for generations.

Amyotrophic lateral sclerosis. ALS.

In his family, the disease was not a rumor or a distant medical abstraction. It had passed through the years like a shadow, taking relatives, stealing movement, narrowing futures. The mutation was in a protein called TDP-43, and by the time Giovanni heard his own diagnosis, his family understood the pattern too well. They knew what progression could look like. They knew how quickly time could disappear.

And they knew there was no treatment waiting for him.

So they asked for the impossible.

Not eventually. Not someday. Now.

Their appeal reached a team of scientists who understood that rare disease does not move on institutional timelines. Biology does not pause for committee review, commercial planning, licensing negotiations, or professional ego. ALS progresses whether the world is ready or not.

In October 2025, Giovanni's patient cells arrived at La Jolla Labs.

There his disease was no longer only a family tragedy. It became a model. A system. A problem that could be studied, measured, and attacked. Illumina performed genome sequencing and phasing, mapping the pathogenic mutation onto a disease-causing haplotype. That map gave the scientists their opening: design a therapy that could selectively target the mutant copy while preserving the normal one.

At La Jolla Labs, scientists designed a series of antisense oligonucleotide candidates, or ASOs, engineered to bind RNA and knock down the mutant genetic signal. Integrated DNA Technologies, part of Danaher, synthesized the screening molecules and delivered them rapidly so the experiment could begin in Giovanni's own patient-derived cellular models.

The screen worked.

Five lead compounds showed strong selectivity in a dose-responsive manner. By late October 2025, one molecule stood out. It could alter the disease-related genetic signal and appeared relatively safe in other cell-based screens. By late November, the lead compound was ready to begin a preclinical safety path.

Giovanni and Francesca his Fiancé named the drug:

GBFsen.

GBFsen was not built as a product chasing a market. It was built as a response to urgency. It was a therapeutic candidate made for a single patient, with no obvious commercial incentive, and yet an extraordinary network of people and companies chose to help.

Illumina. La Jolla Labs. Danaher. ChemGenes. N1C. Thermo Fisher Scientific.

Each contributed expertise, resources, speed, or belief.

And still, the hardest barriers were not always scientific.

In a disease like ALS, every pause has weight.

In early 2026 GBFsen entered rodent toxicology studies. While some parts of drug development can be accelerated; others cannot. A thirteen-week safety study still takes thirteen weeks.

But Giovanni did not have thirteen weeks to spare.

So the team made a careful and thoughtful decision. Clinical dosing would be allowed to proceed after safe interim toxicology reports, while the full preclinical toxicology program continued in parallel. The path forward was conditional on both the animal safety data and Giovanni's own clinical data.

It was urgent. Giovanni was dosed.

The first administration of GBFsen marked a turning point. A drug that had begun as a sequence, a mutation, a family's plea, and a set of patient cells had reached the clinic. It had crossed the distance from genetic diagnosis to human dosing in a matter of months.

For Giovanni and his family, it was not an abstract triumph. It was not a headline. It was hope.



The ability to design and develop GBFsen began with Dr. Har Gobind Khorana, who helped reveal how nucleic acid sequence encodes biological meaning and showed that synthetic DNA and RNA could be used as precise tools. It continued through the work of Bob Letsinger and V. Mahadevan, who helped transform nucleic acid synthesis through solid-phase chemistry, making the construction of defined sequences more practical. It advanced further through Marvin Caruthers, whose phosphoramidite chemistry made the precise synthesis of DNA and RNA routine enough to support the scale required for modern biotechnology. It was strengthened by Fritz Eckstein and colleagues, whose work on phosphorothioate backbone chemistry showed that nucleic acids could be chemically modified to improve stability while preserving biological function. And it became therapeutically actionable through the science of Dr. Joseph Walder, who helped establish the antisense logic that sequence-specific oligonucleotides could inhibit gene expression by binding RNA and recruiting RNase H.